John Doe

Vancouver, BC | anon@example.com | github.com | linkedin.com | Canadian Citizen · Open to Relocation

Education

University of British Columbia April 2028
Bachelor of Science, Computer Science, Option in Software Engineering *GPA: 3.88/4*
Teaching Assistant for Introduction to Computer Systems for 400+ students · Undergraduate TA Award

Experience

Firmware Developer Intern May 2026 – Present
Small Networking Gadget Co. *Vancouver, BC*

- Cut firmware RAM by 400MB and eMMC footprint by 600MB, enough to eliminate a planned memory upgrade by replacing a legacy WebKit UI with a UI implemented in C and LVGL; **projected 2 million dollar hardware upgrade avoided**
- Eliminated the need to flash to hardware test cycle for most UI changes with a host-side build target that runs the firmware UI on a dev machine, mocking the backend and replacing the on-device fbdev, libinput and evdev backend with SDL2
- Improved crash logs by replacing GCC with clang in the debug UI build, enabling support for AddressSanitizer and other sanitizers for rich memory error and undefined behavior error information
- Fixed display tearing by tracing an unconnected line on the schematic, exposing it as GPIO in the device tree, and replacing sequential pixel writes with DMA in the Qualcomm EBI2 driver to eliminate scanline races

Research Intern May 2025 – September 2025
University of British Columbia *Vancouver, BC*

- Achieved a median speedup of 4.97x on large, open-source codebases with long-running integration tests by building a unit test generation tool targeting pytest using Python using dynamic execution tracing, AST transformations, and serialization
- Identified correct assertion generation, not execution capture, as the key barrier to automating carving on real test suites
- Integrated an LLM as a dependency classifier to auto-generate mocks for creating correct and deterministic unit tests

Software Developer Intern September 2024 – April 2025
Mining Co. *Vancouver, BC*

- Eliminated ~100 hours of manual data entry by building a Dataverse-API ingest tool in C#
- Transformed a malformed SQL Server database with multivalued records stored as CSV strings in single columns into actionable business and safety insights, using T-SQL to parse the columns into separate columns and normalized tables
- Introduced the team to Playwright + NUnit to automate end-to-end testing, catching 87% of bugs before manual QA
- Deployed a React.js-based calendar component across all Power Apps teams, implemented in TypeScript

Projects

macOS iSCSI Driver | *C++, DriverKit, IOKit, Swift* November 2025 – Present

- Building an open-source macOS iSCSI initiator that mounts remote storage as a local disk, providing a free, auditable alternative to $249/license proprietary tools
- Bridging the Swift userspace networking client to the driver extension over a lock-free shared-memory ring buffer, saturating 2 10Gb SFP+ NICs, working around DriverKit's lack of network access
- Exposing the remote target as a local block device via the macOS IOUserSCSIParallelInterfaceController interface

Racket Compiler | *x86_64 Assembly, LLVM IR, gdb, Racket* January 2026 – May 2026

- Cut compile times 180x (3h to 1m) by eliminating a pathological typed/untyped bottleneck in Typed Racket
- Replaced x86-only codegen with an LLVM IR backend, enabling compilation to any LLVM-supported platform
- Implemented closure conversion, tagged value representation, and tail call optimization via jump conversion

Sasquatch | *Swift, ARKit, LiDAR, Detectron2, Open3D, Python, FastAPI, PostgreSQL, GCP* March 2026 (24h Hackathon)

- Deployed an ML pipeline combining Detectron2 instance segmentation and LiDAR point clouds to detect climbing wall holds
- Implemented PCA-based wall orientation detection to project 3D point clouds to 2D for inference
- **Reduced false positive rate from 90% to 1%** by adding a Gemini validation pass to filter non-hold wall objects

Skills

Languages: C, C++, Python, Rust, TypeScript, Swift, Java, C#, SQL, Racket
Technologies: LVGL, React, Linux Kernel, OpenWRT, LLVM IR, DriverKit, IOKit
Cloud/Infra: AWS (EC2, ECS), GCP (Cloud Run, Cloud SQL, Gemini), Cloudflare (DNS, Pages), Docker, Linux
Tools/Testing: Git, GitHub Actions, CMake, GDB, pytest, JUnit